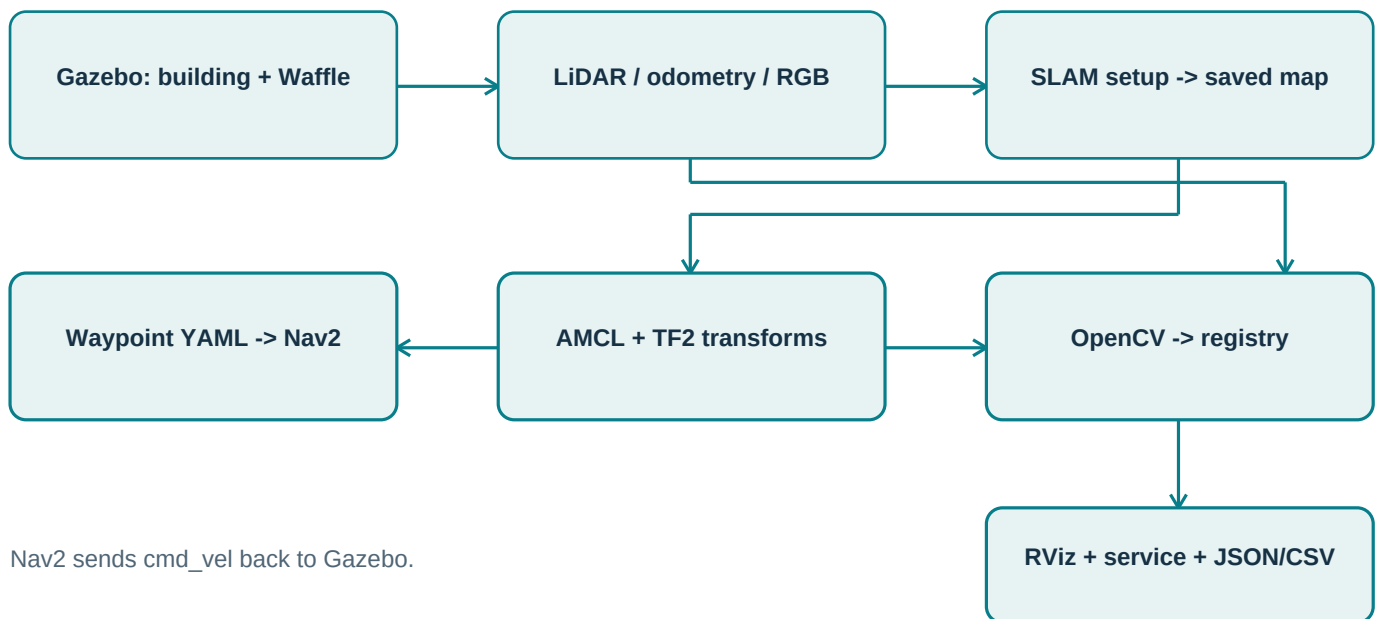


System architecture

Objective

Patrol a four-room simulated building, detect four red cylinders using the onboard RGB camera, and return a deduplicated registry of their global map coordinates. All runtime components execute inside Docker.



Nav2 sends cmd_vel back to Gazebo.

Two separate operating phases

Setup: SLAM Toolbox receives LiDAR scans and odometry while an automated survey visits the rooms. Nav2 map_saver saves the PGM/YAML map. Mission: AMCL localizes on that map and Nav2 executes eight configured goals. The two map-to-odom publishers never run together.

Sensor and motion wiring

The official Waffle model supplies differential drive, wheel joint states, LiDAR and an RGB camera. Robot State Publisher provides fixed link transforms. Camera images and CameraInfo share camera_rgb_optical_frame. The RGB stream is 640 x 480 at a nominal 5 Hz in simulation.

Runtime evidence

The patrol records actual Nav2 action results. The registry exports live JSON/CSV, and a verifier checks map hashes, eight successful goals, four unique victims and position error below 0.5 m. Reference assets and example results are never accepted as live evidence.

Geometry and deduplication

Camera-to-map transform chain

map -> odom -> base_footprint -> base_link -> camera_link -> camera_rgb_frame -> camera_rgb_optical_frame. AMCL supplies map-to-odom; the drive plugin supplies odom-to-base; Robot State Publisher supplies the remaining transforms.

Image-time projection

Optical axes are x right, y down and z forward. HSV thresholding and contours identify red cylinder silhouettes. The two horizontal edges define tangent angles using live focal length and image centre. With radius $r = 0.13$ m, range $= r / \sin(\text{half angular width})$. This estimates the cylinder centre rather than assuming a fixed distance.

Timing and calibration

The observation is stamped with the image time, then TF2 transforms the camera point into the map frame at that time. Missing transforms cause a skipped observation. The detector uses live CameraInfo; the supplied calibration YAML documents the ideal simulated camera. Ground truth is used only by the separate verifier.

Registry rules

Color cylinders have no encoded identity. Anonymous observations receive registry IDs in discovery order. A point within 0.45 m of an existing centroid updates that track using a running mean; otherwise it creates a new track. Three observations are needed before confirmation. Positive source IDs are also supported; invalid numeric values are rejected.

Visible outputs

Confirmed tracks appear as labelled cylinders in RViz and in the /victim_registry service response. JSON and CSV contain stable IDs, map x/y coordinates and a heuristic contour confidence. Files are replaced atomically. Starting a fresh registry resets the run rather than loading old detections.

Tradeoffs and validation limits

This controlled simulation assumes known cylinder size and distinct victims separated beyond the merge radius. Occlusion, other red objects and wrong dimensions can bias results. The setup survey is deterministic, not general exploration. Source tests have passed; a complete ROS2 build and live SLAM/Nav2 verification remain blocked by insufficient host disk space at the time of this document.